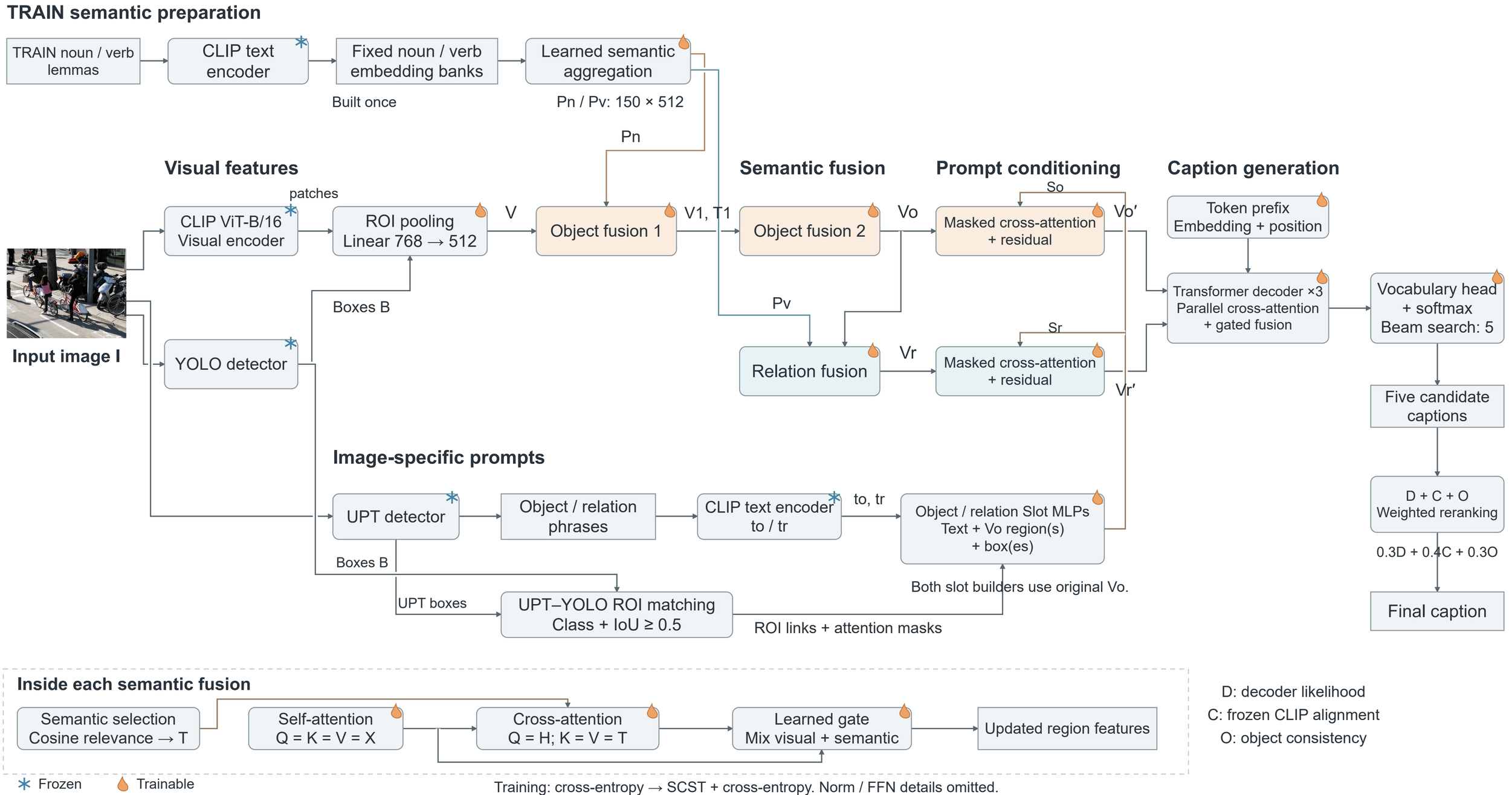

TRAIN semantic preparation
TRAIN noun / verb lemmas
CLIP text encoder
Fixed noun / verb embedding banks
Built once
Learned semantic aggregation
Pn / Pv: 150 × 512
Pn
Visual features
patches
CLIP ViT-B/16 Visual encoder
ROI pooling Linear 768 → 512
V
Boxes B
Input image I
YOLO detector
Semantic fusion
Object fusion 1
V1, T1
Object fusion 2
Vo
Pv
Relation fusion
Vr
Prompt conditioning
So
Masked cross-attention + residual
Vo′
Sr
Masked cross-attention + residual
Vr′
Caption generation
Token prefix Embedding + position
Transformer decoder ×3 Parallel cross-attention + gated fusion
Vocabulary head + softmax Beam search: 5
Five candidate captions
D + C + O Weighted reranking
0.3D + 0.4C + 0.3O
Final caption
Image-specific prompts
UPT detector
Object / relation phrases
CLIP text encoder to / tr
to, tr
Object / relation Slot MLPs Text + Vo region(s) + box(es)
Boxes B
UPT boxes
UPT–YOLO ROI matching Class + IoU ≥ 0.5
ROI links + attention masks
Both slot builders use original Vo.
Inside each semantic fusion
Semantic selection Cosine relevance → T
Self-attention Q = K = V = X
Cross-attention Q = H; K = V = T
Learned gate Mix visual + semantic
Updated region features
D: decoder likelihood
C: frozen CLIP alignment
O: object consistency
Frozen
Trainable
Training: cross-entropy → SCST + cross-entropy. Norm / FFN details omitted.